

Homework 1

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In how many ways can 15 people be seated at a round table if B refuses to sit next to A? What if B only refuses to sit on A's right?

a.

Place a first (1 choice, circular permutation). Then you have 12 choices to seat b (not left of a / right of a / place of a). Then you place any other person, have 13 spots to place (not place of a/b , no other restrictions on placement). By this logic the next person has 12 spots, then the next 11, etc.

$$1 * 12 * 13 * 12 * 11 * \dots = \boxed{12 * 13!}$$

b.

Same steps as part a, except that b has one more spot to sit (to the left of a):

$$1 * 13 * 13! = \boxed{13 * 13!}$$

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Prove that

$$\binom{n}{r} = \binom{n}{n-r}$$

by using a combinatorial argument and not the values of these numbers as given in Theorem 3.3.1.

Let $S = \{\dots\}$, $|S| = n$. Then by definition the number of r -combinations of S is

$$\begin{aligned} |X| &= \{A \subseteq S \mid |A| = r\} \\ |X| &= \binom{n}{r} \end{aligned}$$

We can define the complement of X :

$$\begin{aligned} |A| &= |S| - |\bar{A}| = n - r \\ \bar{X} &= \{B \in S \mid |B| = n - r\} \\ |\bar{X}| &= \binom{n}{n-r} \end{aligned}$$

Therefore, we must show that $|X| = |\bar{X}|$.

To do this, we define f such that $f : X \rightarrow \bar{X}$:

$$\forall A \in X, f(A) = \bar{A} \in |\bar{X}|.$$

f is a bijection (every set has exactly one complement). It follows that

$$|X| = |\bar{X}|$$

$$\binom{n}{r} = \binom{n}{n-r}$$

□

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A group of mn people are to be arranged into m teams each with n players.

a.

Determine the number of ways if each team has a different name.

Let the multiset $M = \{n * t_1, n * t_2, \dots, n * t_m\}$ represent the set of teams. All mn -permutations of this set are:

$$\frac{(nm)!}{n!n!n!n!\dots} = \boxed{\frac{(nm)!}{(n!)^m}}$$

b.

Determine the number of ways if the teams don't have names.

In this case the previous result is overcounting by $m!$, since all permutations of different team labels are counted separately. Since all the teams are of the same size, the size of all the partitions are the same as well, meaning we can use the division principle to get:

$$\boxed{\frac{(nm)!}{m!(n!)^m}}$$

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Determine the number of 10-permutations of the multiset $S = \{3 \text{ a}, 4 \text{ b}, 5 \text{ c}\}$.*

Instead of constructing the permutation we can remove 2 elements from S and count the 10-permutations of that new multiset. This must be done for every combination of item removed:

$$\begin{aligned}
(a, a) &\rightarrow S = \{1 * a, 4 * b, 5 * c\} \rightarrow \frac{10!}{1!4!5!} \\
(a, b) &\rightarrow S = \{2 * a, 3 * b, 5 * c\} \rightarrow \frac{10!}{2!3!5!} \\
(a, c) &\rightarrow S = \{2 * a, 4 * b, 4 * c\} \rightarrow \frac{10!}{2!4!4!} \\
(b, b) &\rightarrow S = \{3 * a, 2 * b, 5 * c\} \rightarrow \frac{10!}{3!2!5!} \\
(b, c) &\rightarrow S = \{3 * a, 3 * b, 4 * c\} \rightarrow \frac{10!}{3!3!4!} \\
(c, c) &\rightarrow S = \{3 * a, 4 * b, 3 * c\} \rightarrow \frac{10!}{3!4!3!}
\end{aligned}$$

$$|P| = |P_{a,a}| + |P_{a,b}| + \dots$$

$$|P| = \frac{10!}{1!4!5!} + \frac{10!}{2!3!5!} + \frac{10!}{2!4!4!} + \frac{10!}{3!2!5!} + \frac{10!}{3!3!4!} + \frac{10!}{3!4!3!}$$

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Determine the total number of combinations (of any size) of a multiset of objects of k different types with finite repetition numbers $n_1, n_2, \dots, n_k$, respectively.

For every element we can choose in the range $[0, n_k]$ number of objects for the combination. This means the # of choices for each element is $n_k + 1$. We can use the product rule to multiply the choices for each element since the choice happens for every element for every combination.

$$(n_1 + 1) * (n_2 + 1) * \dots * (n_k + 1)$$

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How many integral solutions of

$$x_1 + x_2 + x_3 + x_4 = 30$$

satisfy $x_1 \geq 2, x_2 \geq 0, x_3 \geq -5, x_4 \geq 8$?

$$y_1 = x_1 - 2, y_2 = x_2, y_3 = x_3 + 5, y_4 = x_4 - 8$$

$$y_1 + y_2 + y_3 + y_4 = 30 - 2 + 5 - 8$$

$$y_1 + y_2 + y_3 + y_4 = 25$$

$$y_1, y_2, y_3, y_4 \geq 0$$

This becomes the same as combinations of a multiset with infinite repetition numbers:

$$\begin{aligned}
\binom{k+r-1}{r} &= \binom{4+25-1}{25} \\
\binom{28}{25} &= \frac{28!}{25!(28-25)!} = \boxed{\frac{28!}{25!3!}}
\end{aligned}$$

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There are n sticks lined up in a row, and k of them are to be chosen.

- How many choices are there?
- How many choices are there if no two of the chosen sticks can be consecutive?
- How many choices are there if there must be at least l sticks between each pair of chosen sticks

A

n sticks to choose from, k to be chosen:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

B

We can imagine first lining up $n - k$ spots of non-chosen places (whose sticks are unknown yet). Then, we insert k sticks between those non-chosen places. After placing all chosen sticks, the positions of chosen/non-chosen places will be valid (no consecutive chosen sticks).

Therefore the answer becomes equal to number of ways to insert (select the between points). There are $n - k$ non-chosen spots, therefore $n - k + 1$ gaps where to insert:

$$\boxed{\binom{n-k+1}{k}}$$

C

Instead of placing all the chosen spots, we can first place all the chosen spots as well as the gaps between them, then insert non-chosen spots between them.

The smallest sequence satisfying the l sticks condition is:

$$\overbrace{c \underbrace{nnnn}_{l \text{ sticks}} c \underbrace{nnnn}_{l \text{ sticks}} \dots c}^{k+l*(k-1) \text{ in length}}$$

Therefore, the length of this sequence is $k + l * (k - 1)$.

Now we must insert a variable number of sticks between this fixed sequence. The number of sticks inserted at each position varies, but the total number of inserted sticks remains the same (what's left over from the fixed sequence).

The number of place where sticks can be inserted is $k + 1 \rightarrow$ between chosen($k - 1$), to the left of the leftmost chosen(1), or to the right of the rightmost chosen(1). Therefore, the combinations can be modeled by:

$$\begin{aligned} x_1 + x_2 + x_3 + \dots + x_{k+1} &= n - (k + l * (k - 1)) \\ x_1 \geq 0, x_2 \geq 0, \dots, x_{k+1} &\geq 0 \end{aligned}$$

With the stars and bars principle the number of positive integral solutions can be found:

$$\begin{aligned}
& \binom{(k+1) + (n - (k + l * (k-1))) - 1}{(k+1) - 1} \\
& \rightarrow k + 1 + n - (k + l * (k-1)) - 1 \\
& \rightarrow k + n - (k + l * (k-1)) \\
& \rightarrow k + n - k - l * (k-1) \\
& \rightarrow n - l * (k-1) \\
& \boxed{\binom{n - l * (k-1)}{k}}
\end{aligned}$$

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Determine the number of ways to distribute 10 orange drinks, 1 lemon drink, and 1 lime drink to four thirsty students so that each student gets at least one drink, and the lemon and lime drinks go to different students.

Let the set of valid combinations be S . We can partition S into 3 distinct cases:

1. all students get at least 1 orange
2. one student gets no oranges
3. two students get no oranges

The case with three students with no oranges contains no solutions (after distributing lime and lemon there is still one person with nothing). So it is not considered

Case 1

Lets first distribute all oranges, then distribute the lemon and lime.

ways to distribute oranges such that everyone gets at least one:

$$\begin{aligned}
x_1 + x_2 + x_3 + x_4 &= 10 \\
x_1, x_2, x_3, x_4 &\geq 1 \\
&\rightarrow \\
y_1 = x_1 - 1, y_2 = x_2 - 1, \dots, y_4 = x_4 - 1 \\
&\rightarrow y_1 + y_2 + y_3 + y_4 = 10 - 4 \\
y_1 + y_2 + y_3 + y_4 &= 6 \\
y_1, y_2, y_3, y_4 &\geq 0
\end{aligned}$$

The number of integral solutions becomes:

$$\binom{n+k-1}{n} \rightarrow \binom{6+4-1}{6} \rightarrow \binom{9}{6}$$

For each of these choices there are no restriction on placement of lime or lemon other than them not going to the same person, so the choices for their placement is:

$$4 * 3$$

Thus, all valid combinations for at least one orange distributed among everyone is:

$$\boxed{\binom{9}{6} * 4 * 3}$$

Case 2

Now the second case, where one person does not get any oranges is solved similarly.

We can consider the number of ways to distribute among two people, then permute all those among all possible combinations of people:

$$\begin{aligned} x_1 + x_2 + x_3 &= 10, x_1, x_2, x_3 \geq 1 \\ \rightarrow y_1 + y_2 + y_3 &= 7, y_1, y_2, y_3 \geq 0 \end{aligned}$$

$$\binom{7+3-1}{7} \rightarrow \binom{9}{7}$$

we must choose who does not get any oranges:

$$\binom{4}{1} = 4$$

So the combinations without including the lemon or lime is:

$$\binom{9}{7} * 4$$

Now to insert the lemon and lime let's find all distributions that must include the person with nothing:

We pick anyone in the group, then give them either lime or lemon. Then, we give the other to the person with nothing. Therefore the combinations are:

$$3 * 2$$

Since there are 3 people that are not the person with nothing, and 2 ways to permute lemon and lime.

So the total combinations are:

$$\boxed{\binom{9}{7} * 4 * 3 * 2}$$

Case 3

In this case we must distribute oranges among 2 people:

$$\begin{aligned} x_1 + x_2 &= 10, x_1, x_2 \geq 1 \\ \rightarrow y_1 + y_2 &= 8, y_1, y_2 \geq 0 \end{aligned}$$

$$\binom{8+2-1}{8} \rightarrow \binom{9}{8}$$

we must choose which people get the oranges:

$$\binom{4}{2} = \frac{4!}{2!2!} = \frac{4 * 3 * 2}{2 * 2} = 3 * 2$$

So the combinations for distributing oranges is:

$$\binom{9}{8} * 3 * 2$$

Now for placing the lemon and lime we must give them to the people that have nothing, so the choices are 2, since who gets lime or lemon doesn't matter. Thus, the combinations for this case are:

$$\binom{9}{8} * 3 * 2 * 2 = \boxed{\binom{9}{8} * 3 * 4}$$

Result

We can add all these combinations since they are mutually exclusive partitions of the solution set, therefore the total number of combinations is:

$$\boxed{\binom{9}{6} * 4 * 3 + \binom{9}{7} * 4 * 3 * 2 + \binom{9}{8} * 3 * 4}$$

48

Prove that the number of permutations of m A's and at most n B's equals

$$\binom{m+n+1}{m+1}$$

Lets first split all permutations by their size:

m -permutations: $M = \{m * A\} \rightarrow 1$ way

$m + 1$ -permutations: $M = \{m * A, 1 * B\}$

$m + 2$ -permutations: $M = \{m * A, 2 * B\}$

...

$m + n$ -permutations: $M = \{m * A, n * B\}$

Finding the size of each size-permutation becomes:

$$\begin{aligned} m &\rightarrow \boxed{\binom{m}{m} = 1} \\ m + 1 &\rightarrow \binom{m+1}{1} = \boxed{\binom{m+1}{m}} \\ m + 2 &\rightarrow \binom{m+2}{2} = \boxed{\binom{m+2}{m}} \\ &\dots \\ m + n &\rightarrow \binom{m+n}{n} = \boxed{\binom{m+n}{m}} \end{aligned}$$

Therefore, the answer is equivalent to:

$$\binom{m}{m} + \binom{m+1}{m} + \binom{m+2}{m} + \dots + \binom{m+n}{m}$$

or

$$\binom{m+n}{m} + \binom{m+n-1}{m} + \binom{m+n-2}{m} + \dots + \binom{m}{m}$$

So we must now prove equivalence of

$$\binom{m+n+1}{m+1} = \binom{m+n}{m} + \binom{m+n-1}{m} + \binom{m+n-2}{m} + \dots + \binom{m}{m}$$

To prove this pascal's identity is recursively expanded, giving the same sum:

$$\begin{aligned} \binom{m+n+1}{m+1} &= \binom{m+n}{m+1} + \binom{m+n}{m} \\ &= \left(\binom{m+n-1}{m+1} + \binom{m+n-1}{m} \right) + \binom{m+n}{m} \\ &= \left(\left(\binom{m+n-2}{m+1} + \binom{m+n-2}{m} \right) + \binom{m+n-1}{m} \right) + \binom{m+n}{m} \\ &\quad \dots \\ \binom{m+n+1}{m+1} &= \binom{m+2}{m+1} + \dots + \binom{m+n-2}{m} + \binom{m+n-1}{m} + \binom{m+n}{m} \\ &\rightarrow \binom{m+2}{m+1} = \binom{m+1}{m+1} + \binom{m+1}{m} \\ &\quad \binom{m+2}{m+1} = \binom{m}{m} + \binom{m+1}{m} \\ &\rightarrow \boxed{\binom{m+n+1}{m+1} = \binom{m}{m} + \binom{m+1}{m} + \dots + \binom{m+n-2}{m} + \binom{m+n-1}{m} + \binom{m+n}{m}} \end{aligned}$$

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Prove that the number of permutations of at most m A's and at most n B's equals

$$\binom{m+n+2}{m+1} - 1$$

We can first fix the # of A's and find the permutations having at most n B's.

From problem #48 the number of permutations of m A's and at most n B's equals:

$$\binom{m+n+1}{m+1}$$

In this case we sum over all different numbers of A, yielding the sum:

$$\sum_{k=0}^m \binom{k+n+1}{k+1}$$

This sum can be simplified using the same equality found from question #48:

$$\binom{m+n+1}{m+1} = \sum_{k=0}^n \binom{m+k}{m} = \binom{m}{m} + \binom{m+1}{m} + \cdots + \binom{m+n}{m}$$

However, to use this equality the summation has to be transformed:

$$\begin{aligned} &= \sum_{k=0}^m \binom{(k+1)+n}{(k+1)} \\ &= \sum_{k=0}^m \binom{n+(k+1)}{n} \\ &= \sum_{k=1}^{m+1} \binom{n+k}{n} \\ &= \sum_{k=0}^{m+1} \binom{n+k}{n} - \binom{n+0}{n} \\ &= \sum_{k=0}^{m+1} \binom{n+k}{n} - 1 \\ \rightarrow \sum_{k=0}^{m+1} \binom{n+k}{n} - 1 &= \binom{n+(m+1)+1}{n+1} - 1 \\ &= \boxed{\binom{n+m+2}{m+1} - 1} \end{aligned}$$

51

Consider the multiset $M = \{n \text{ a}, 1, 2, 3, \dots, n\}$ of size $2n$. Determine the number of its n -combinations.*

We can fix the amount of a 's and count each of those permutations. For each non- a we are choosing from the $\{1, 2, 3, \dots, n\}$ elements:

$$\begin{aligned} \{n * a\} &\rightarrow 1 = \binom{n}{0} \\ \{(n-1) * a, ?\} &\rightarrow \binom{n}{1} \\ \{(n-2) * a, ?, ?\} &\rightarrow \binom{n}{2} \\ &\dots \\ \{0 * a, 1, 2, 3, \dots, n\} &\rightarrow \binom{n}{n} \end{aligned}$$

Leading to

$$\boxed{\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = \sum_{k=0}^n \binom{n}{k} = 2^n}$$